



Somewhere between acceptable and sustainable: When do impacts to resources become too large in protected areas?

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ARTICLE INFO

Keywords:

Protected areas
Impacts
Values
Acceptability
Ship strikes
Glacier Bay National Park

ABSTRACT

Utilization of marine and terrestrial protected areas is fundamentally important for their acceptance and success. Yet even appropriate uses can negatively impact resources requiring managers to make decisions as to when the impacts become unacceptably large. These decisions can be difficult because the level at which impacts occur may be far below the level at which resource persistence is threatened. In Glacier Bay National Park, Alaska, managers must make a recurring decision regarding the number of cruise ships that are allowed to enter the park each year. Cruise ships bring > 95% of all visitors to the park but have been involved in several lethal collisions (ship strikes) with humpback whales. Using an individual-based simulation model, we demonstrate that increasing the annual ship volume from current to maximum allowable levels would have negligible impacts on population growth of whales. Over the next 30 years the median number of collisions would likely increase from 3 (95% CI: 0–7) to 4 (1–8) or, worst case scenario, from 5 (0–7) to 8 (3–13), while median annual growth rates would, at most, shift from 4.4% (3.7%–5.2%) to 4.2% (3.5%–4.9%), depending upon assumptions. By comparison, a median of 67 (50–82) ship strikes would need to occur over the next 30 years to threaten the persistence of whales. Confronted with an impact level that is far below what would threaten the conservation of whales, managers are tasked with the decision of placing values on 2 million additional visitors for every additional dead whale. We argue that decision-making related to use-impact trade-offs for protected areas could be more consistent and effective if site-values are defined explicitly, clearly communicated among stakeholders, and linked to biological metrics. Protected areas managers can then utilize monitoring programs to evaluate management effectiveness when the objective is conserving both resources and values.

1. Introduction

Reserves, parks, conservation zones, and other types of protected areas around the world are established primarily to conserve natural resources considered to be ecologically, historically, or culturally important. Both internal and external stressors can negatively impact resources within these areas (Thurstan et al., 2012; Hansen et al., 2014), but the magnitude of those impacts, and thus the need for management action, varies considerably. In some cases, such as when an existential threat develops that would lead to permanent loss of large areas of habitat and/or extirpation of biodiversity (e.g., damming a river in a protected area; e.g., Laranjeiras et al., 2014) the need for management action is obvious because it represents a clear violation of the goals and objectives of the protected area. More frequently, however, impacts are sufficiently limited such that the need for management action is equivocal. Confounding management uncertainty, these impacts may derive

from appropriate, and perhaps necessary, uses of the protected area.

For example, visitation (and visitor activities) is one of the most common sources of direct and indirect impact to resources in protected areas. In the U.S. visitation to national parks has been linked to wildlife disturbance (e.g., Coleman et al., 2013; Young et al., 2014), incidental deaths of wildlife (e.g., animal-vehicle collisions; Ament et al., 2008), degradation of air or water quality from pollutants or human waste (e.g., Reed and Rasnake, 2016), noise pollution (e.g., Kaplan and Mooney, 2015), and destruction of fish or wildlife habitat (e.g., Hallac et al., 2012), among others. Yet these impacts may be highly localized, temporary, involve only a small fraction of animals or sites within the area, or produce conditions nearly identical to background (unimpacted) levels (Martin et al., 2009; Clow et al., 2011; Mölders and Gende, 2015; Marcella et al., 2016; Eagleston and Marion, 2017). What's more, providing opportunities for visitor use and enjoyment of park resources represents a fundamental management goal of the

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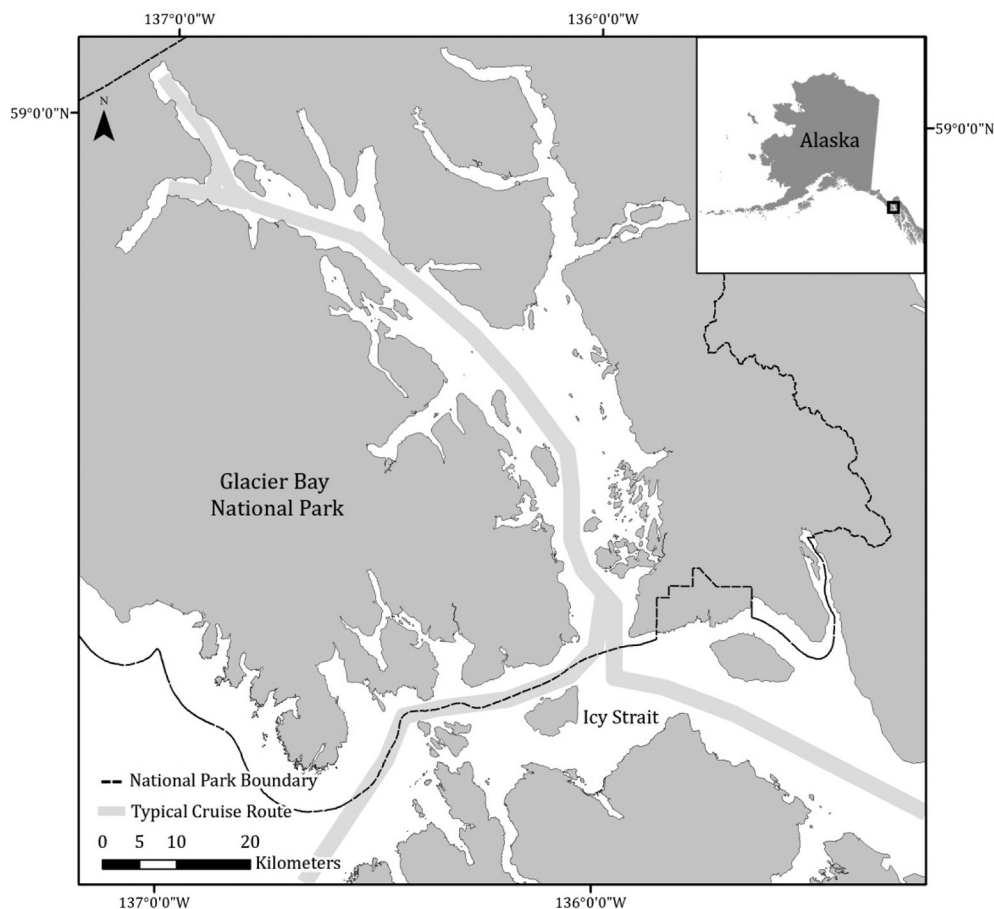


Fig. 1. Glacier Bay National Park and adjacent Icy Strait showing the area used by cruise ships when entering, within, and exiting park waters (light shaded area). Ship routes rarely deviate from outside of this area.

National Park Service (NPS, 2006; p 10), mirroring the goals of parks in other countries or World Heritage Sites. These conflicting issues also commonly apply to marine protected areas because management success is often contingent upon support from local communities whose members may rely on extractive uses of the area, such as for subsistence and artisanal fishing (Guidetti and Claudet, 2010). The conflicting incentives complicate the decision-making process because, absent clear definitions or guidance, managers must decide when the impacts from mandated and important uses of the area become unacceptably large.

As an applied example, we consider the tradeoffs between cruise ship traffic and the associated impacts to humpback whales (*Megaptera novaeangliae*) in Glacier Bay National Park, Alaska (hereafter ‘Glacier Bay’ or ‘the park’). Glacier Bay is one of the largest marine protected areas in the northern hemisphere, and large cruise ships ($\bar{x}$ length of ships visiting in 2012 = 273 m; Webb and Gende, 2015) have been visiting the area regularly since the early 1970s. While consistently conveying > 95% of the visitors to the park annually, cruise ships, like other marine vessels, can negatively impact marine wildlife (Young et al., 2014; Marcella et al., 2016), including humpback whales that aggregate within the park and adjacent waters to feed during the summer months (Saracco et al., 2013). Ship routes overlap temporally and spatially with heavy whale-use areas (Webb and Gende, 2015) resulting in frequent surfacing events near the bow of the transiting ships (Gende et al., 2011; Harris et al., 2012; Williams et al., 2016). Consequently, collisions between ships and whales (ship strikes) have been documented within the park (Neilson et al., 2012), despite speed restrictions on vessels where whales aggregate and requirements for vessels to maintain a mid-channel course to minimize overlap with near-shore habitat used by whales. While relatively rare, ship strikes have been demonstrated to negatively impact the dynamics of some

populations of whales (Fujiwara and Caswell, 2001) particularly when they involve adult females because the collision not only removes her from the population but also results in a loss of her future offspring in the area owing to high levels of maternally-driven site fidelity (Baker et al., 2013; Pierszalowski et al., 2016). The relationship between cruise ship volume and risk of whale deaths (in addition to risk of impacts to other resources such as to air quality from ship pollution) forces managers to make decisions regarding appropriate tradeoffs between increased visitation and the associated probability that an increased number of whales will be killed.

While a dead whale washing up on the shore of a national park is clearly undesirable, the fundamental aspect of this issue reflects the common and recurring issue faced by nearly all protected area managers: at what level does an impact from an appropriate use of the area become unacceptably large? An obvious first approach, such as that commonly used in evaluating by-catch in fisheries (Underwood et al., 2008) is to ask whether the impact threatens the persistence of the resource. If so, the need for management intervention is clear. However, should the impact level be deemed biologically sustainable, managers are still tasked with deciding if and when the impact level becomes sufficiently large that use or visitation needs to be curtailed. The point at which management actions are invoked is important to define and communicate because conflict can arise from stakeholders who may not understand the criteria upon which decisions are made (e.g., Borkowski et al., 2006), or adhere to different values about how protected areas should be managed (Sarewitz, 2004) particularly if management is based on factors other than sustainability. Ambiguity can also lead to loss of support for the conservation areas (DiFranco et al., 2016).

Here we quantify the potential impacts of changes in cruise ship

traffic to both visitor levels and the population of humpback whales in and near Glacier Bay through a series of simulations. We quantify how a decision to increase visitation levels may influence the number of ship strikes and, by extension, the population growth rate of whales in the park. We then compare this level of ship strikes to a conservative estimate of the number of ship strikes that the population could sustain, i.e., that would need to occur to threaten the persistence of whales using the park. We use our findings to highlight how, absent risk to biological sustainability, management actions effectively shift from conserving resources, based on arguments of biological sustainability, to conserving values. We discuss how the development and identification of explicit definitions of protected area values clearly linked to management objectives will enhance management effectiveness by increasing consistency and transparency and possibly reducing conflict with other stakeholders.

2. Materials and methods

2.1. Study area

Located in the southeastern archipelago of Alaska, Glacier Bay National Park and Preserve (Fig. 1) encapsulates > 2.3 million acres of terrestrial and marine areas. The park's 600,000 ha of marine acreage are unusually productive (Reisdorph and Mathis, 2014) which attracts large aggregations of marine mammals (Mathews et al., 2011) including humpback whales (Saracco et al., 2013). The number of whales using Glacier Bay and adjacent Icy Strait numbered 181 (95% CI: 173.5–196.2) individuals as of 2009 (Saracco et al., 2013) comprising a part of the feeding aggregation (2883–6414 individuals) that encompasses northern British Columbia and southeastern Alaska (Muto et al., 2017). Annual population growth rate for the segment of the population using Glacier Bay-Icy Strait has averaged 4.4% since 1985 (95% CI: 3.0–5.8%; Saracco et al., 2013).

2.2. Cruise ship management in Glacier Bay

The number and operations of all vessels, including cruise ships, are regulated by the NPS according to the Glacier Bay Vessel Quota and Operating Requirements Final Environmental Impact Statement (VQOR EIS; NPS, 2003), which stipulates that a maximum of 2 large cruise ships can enter Glacier Bay per day. However, seasonal quotas (NPS, 2003) are split into a 92-day 'prime' season (1 June–31 August), and a 61-day 'shoulder' season (May and September). Currently seasonal quotas, which are decided annually by the park Superintendent, are 153 and 72 for the prime and shoulder seasons, respectively. The total allowable number of large cruise ship entries into Glacier Bay ($153 + 72 = 225$) is thus currently 81 entries lower than the maxima (2 ships every day for the 153-day May–September season = 306).

2.3. Simulation model

2.3.1. Demographics

We constructed an individual-based simulation model in R (Version 3.1.2; The R Foundation for Statistical Computing; <https://www.r-project.org/>) to understand (1) how the whale population using Glacier Bay may be impacted by changes in cruise ship entries and (2) how this impact level compares with the level necessary to threaten the persistence of the whale population using the park. Our model assigned probabilities to birth, maturation, and survival with individual life-history events modeled as Bernoulli random variables, with class-specific rates defining the likelihood of events. The model tracked the fate of 3 classes including adult males (m), adult females (f), and juveniles (j).

We began simulations in year 2010 corresponding to the year after the last abundance estimate was produced ($N = 181$ whales; Saracco et al., 2013). Simulations were run for 30 years (2010–2039) to allow

for population processes to occur, such as maturation, productivity, and mortality, that would affect the number of whales utilizing park waters at the multi-decadal scale. We modeled initial abundances N_1 of individuals $i = (1, \dots, I)$ in each of the 3 classes using a multinomial distribution, $N_{1,c} \sim \text{Multinom}(p_c, N_1)$, where p_c is the probability of being in class $c = (j, f, m)$.

Annual survival was modeled for each individual in each year $t = (1, \dots, T)$ based on a class-specific natural survival rate of 0.918 for juveniles and 0.984 for adults (Zerbini et al., 2008) and assumed that natural mortality occurred prior to any ship strike events. To facilitate the model calculations, we defined a survival matrix $S (\times T)$ that tracks the survival status of each individual over time, with elements $s_{i,t}$ equal to 1 (alive) or 0 (dead). Survival of each individual was modeled as a Bernoulli event given the class-specific survival rate $s_{i,t} = \text{Bern}(pn_{c,t})$, where $pn_{c,t}$ was the class-specific probability of natural survival.

Each individual was assigned an age (age_i), with adult males and females in the initial population $N_{1,c}$ assigned ages that are > 11 years, and juveniles < 10 years. Maturation of juveniles was modeled as a stochastic process, via $M_i \sim \text{Bern}(pm)$, where the probability of maturation pm is modeled as $pm = \text{logit}^{-1}(2age_i - 8)$ where $\text{logit}^{-1}(x) = \exp(x) / (1 + \exp(x))$ resulting in an increasing probability of maturing to an adult as age increases beyond 6 years (50% probability of maturity at 8 years and 100% at 11 years). We assumed the initial population consisted of 50% juveniles, 25% adult males, and 25% adult females (Polanowski et al., 2014). Juveniles matured into adult females or adult males with equal probability, which was sampled via a Bernoulli random variable with probability 0.5 if ($M_i = 1$) and $c_i = 2 + \text{Bern}(0.5)$; otherwise $c_i = 1$.

Productivity, i.e., the number of calves produced by each mature female annually, was assumed to average 0.375 (Zerbini et al., 2008). For each adult female, the probability of birth was modeled as a Bernoulli event $B_{i,t} = \text{Bern}(pb)$, where pb is the probability of birth and survival from the calving grounds to the winter feeding grounds. All newly born juveniles were then added to the population to allow the population to grow over time, modeled as $N_{c=j,t+1} = \sum B_{i,t}$.

2.3.2. Collision rate and detection probability

Since 1985, the NPS has supported an intensive humpback whale monitoring effort covering Glacier Bay and adjacent waters (Gabriele et al., 2017) during which time a total of 2 ship strikes have been detected in the park. In 2001, a pregnant adult female was struck and killed by a cruise ship at the mouth of Glacier Bay; in 2004 a necropsy of a male humpback whale calf found freshly dead in the park concluded that the injuries were consistent with a probable vessel strike (Neilson et al., 2012). Although the vessel couldn't be identified, we assumed it died as a result of collision with a large cruise ship.

Despite intensive monitoring efforts and relatively small area, it's possible that the number of ship strikes detected represent a subset of those events that have actually occurred. We thus initially defined the collision rate parameter as the total number of detected collisions (2) relative to the total number of ship entries from 1985 to 2009 (4622) equating to a per entry collision probability of 0.000433. However, given the uncertainty in this estimate, we also ran simulations assuming a historical probability of detection (PrD) of 50%, i.e., the true collision rate was double of that detected equating to a per entry collision rate of 4/4622 or 0.000865 per ship entry.

We were thus able to model the varying number of whales struck per year and the impacts to growth rates by either increasing the volume of ships that enter Glacier Bay, or by direct removals. The number of ship strike events in a given year was modeled using a Poisson distribution $Strikes_t \sim \text{Pois}(r_t)$ where r_t is the annual collision rate and is a function of the number of vessels transiting in year t and the per-vessel strike probability. For each future collision $Strikes_t$, we also designated the age/sex as either random or involving only adult females (maximizing impact on the population). If an individual is identified as struck, then the value of $s_{i,t}$ is updated to 0 to reflect the ship strike

mortality event. We assumed all ship strikes resulted in instantaneous mortality and that the population was closed to emigration and immigration to maximize impacts to the 'Glacier Bay population'.

We used the simulation model to contrast the impacts to visitation levels and whale populations should the park continue with existing (BASE) ship quotas (225 total) vs. one that would maximize (MAX) the daily quota of two per day every day during the entire 153-day, May–September tourist season (306 total). We first calculated how many collisions would likely occur over 30 years for BASE vs. MAX under both detection probability scenarios ($PrD = 1$ vs. 0.5). Based on those scenarios, we estimated median population growth rate to explore the population impacts should the additional collisions occur. Simulations were run for a scenario where all future collisions are random with respect to sex/age classes, and a scenario where all future collisions involved adult females to represent a 'worst case estimate' (e.g., Mangel, 1993). We then re-ran the simulations by varying the number of collisions (removals) to explore the level at which ship strikes may threaten the persistence of whales (e.g., Monnahan et al., 2015). Sustainability was defined as the ship strike level at which median population growth remained positive. For these levels, we ran simulations by specifying all future collisions as random with respect to age/sex classes and where all future collisions involve adult females.

Finally, the National Park Service has a dual mandate in conserving park resources and values while also providing opportunities for visitor use and enjoyment. Thus the implications to visitation are a necessary component of any management action. To accurately evaluate how the different management scenarios (BASE vs. MAX) will influence the number of visitors (ship passengers) to Glacier Bay, we utilized data provided to the park each year from the cruise companies on ship capacity and passengers per ship. These metrics change due to the addition of berth space (bunks) within a ship, or by different sizes of ships visiting the park across years. We averaged the total number of passengers aboard each unique ship that entered Glacier Bay since 1995 (when data were first available), and used linear regression to model the trend in average passenger capacity over time. The average capacity was then multiplied by the number of ships entering the park over time to give an accurate estimate of the total number of visitors each year. Details on model structure can be found in Supporting Information.

3. Results

3.1. Ship and passenger volume

From 1995 to 2009, a total of 68 unique cruise ships entered Glacier Bay with the annual number ranging between 15 and 23. Total ship entries varied from 168 (1995) to 225 (2008) with total number of ship passengers varying between 206,000 (1995) and 423,000 (2009). Reflecting the industry's trend in building larger ships and retrofitting existing ships with more berths (bunks), the average number of passengers aboard each ship has significantly increased over time ($\bar{x} \#$ of passengers = $41.1 * \text{year} - 80,641$; $R^2 = 0.79$, $p < 0.0001$; Fig. 2) with the average increasing 57% between 1995 ($\bar{x} = 1175$) and 2009 ($\bar{x} = 1844$) or nearly 4% per year. Thus, if the park increases the annual ship volume from 225 (BASE) to 306 (MAX) per year, and accounting for changes in the average increase in ship capacity, the total number of additional visitors to Glacier Bay over the next 30 years would be 6,044,113 or just over 200,000 per year (Table 1).

3.2. Increased visitation and ship strikes

Assuming all ship strikes prior to 2009 were detected ($PrD = 1$), the median number of collisions over the next 30 years was estimated to be 3 for BASE (225 ships annually; 95% CI: 0–7) and 4 for MAX (306 ships annually; 95% CI: 1–8; Fig. 3a) equating to a single additional ship strike for 2430 additional ship entries. Assuming the historical detection probability for ship strikes was 0.5 rather than 1, the median

number of collisions after 30 years was estimated to be 5 for BASE (95% CI: 2–11) and 8 for MAX (95% CI: 3–13; Fig. 3b). Thus, the latter, worst case scenario, generates a tradeoff for managers of approximately one additional dead whale for every 2 million additional visitors. The former scenario equates to 1 additional dead whale per 6 million additional visitors.

3.3. Impacts of increased visitation on whale population growth rates

Regardless of ship entry levels, or assumptions related to detection probability or demographics of whales involved in ship strikes, our simulation model demonstrated that the number of expected ship strikes does little to affect population growth rate (Fig. 4). Median population growth rates were approximately 4.4% for each scenario where all future collisions were random with respect to age and sex class, regardless of historical detection probability. The biggest impact to growth rates occurred for the scenario where historical probability of detection was 50% and all future collisions involved adult females. In this scenario, median growth rates shifted from 4.4% (95% CI: 3.7%–5.2%) to 4.2% (CI: 3.5%–4.9%). The loss was attributed not just to the number of adult females struck, but also the lost calves that would have returned to Glacier Bay, recruited into the adult population, and produced calves of their own.

By comparison, the ship strike level necessary to affect the sustainability of the population of whales using Glacier Bay was much higher. For example, to shift the median growth rate from 4.4% annually to 0.1% annually, i.e., to a level where no growth occurs, ships would need to be involved in 226 (95% CI: 98–225) lethal collisions with whales over the next 30 years assuming all collisions are random with respect to age and sex classes. This equates to approximately 7.5 whale deaths per year, or a nearly 57-fold rise over the 0.13 per year estimated for MAX ships (when $PrD = 1$). If all future collisions were adult females, a total of 67 (95% CI: 50–82) ship strikes would need to occur (~2.2 per year; Fig. 5).

4. Discussion

Managers of protected areas are increasingly confronted with decisions balancing conservation mandates with high demand for access and unprecedented threats from altered ecosystems in adjacent areas (Hansen et al., 2014). In the case of U.S. National Park Service, policy requires that park management decide whether visitation levels or use will result in unacceptable impacts to park resources. In some cases, when an activity threatens the future existence of a resource and thus the ability for future generations to enjoy it, the decision to limit activity level is straightforward. However, as is the case for our Glacier Bay example, increases in visitor volume or activities may simply increase the frequency of a known impact, but at a level well below that which may threaten persistence of that resource within the protected area. As virtually 'every form of human activity that takes in place within a park has some degree of effect on park resources...' (NPS, 2006) managers are thus tasked with deciding where within the spectrum between small/inconsequential and large/unsustainable that the impact becomes unacceptable. In these instances, managers are effectively valuing individual sites, wildlife, or processes, relative to visitor volume (activities), and thereby shifting the objective from conserving the park's resource to conserving the park's values.

4.1. Glacier Bay visitation vs. ship strikes

In Glacier Bay we found that the abundance of whales in the park and adjacent waters were highly robust to the changes in cruise ship volume. The 81-ship increase per year (MAX) only marginally increased the probability of a collision given the existing low per-entry collision probability, even assuming historical detection probabilities were low. Consequently, the number of additional ship strikes increased by only 1

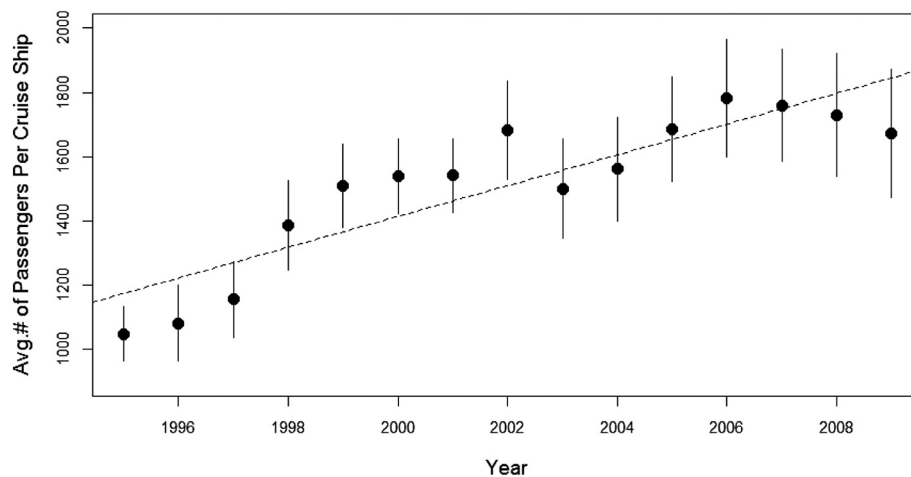


Fig. 2. Average ($\pm$ SE) number of passengers per cruise ship that entered Glacier Bay, 1995–2009. Line represents the best fit linear model: $\bar{x}$ # of passengers = $41.1 * \text{year} - 80,641$; $R^2 = 0.79$, $p < 0.0001$. The total number of different cruise ships that entered the park annually varied between 15 and 23.

Table 1

Projected changes in average number of passengers per cruise ship (in 5-year intervals) entering Glacier Bay, and resulting total number of passengers under a management decision of existing ship entry levels (BASE: 225/year) vs. maximum allowable (MAX: 306/year).

Year	Avg. pax. per ship	# visitors		Difference
		BASE ships	MAX ships	
1	1937	435,760	592,633	156,873
5	2089	469,934	639,110	169,176
10	2278	512,651	697,205	184,554
15	2468	555,368	755,301	199,933
20	2658	598,086	813,397	215,311
25	2848	640,803	871,492	230,689
30	3038	683,521	929,588	246,067
Cummulative		16,789,204	22,833,318	6,044,114

or 3 (depending upon assumptions) after three decades. These losses, even if they were adult females, were more than offset by modest productivity and recruitment of calves into the adult population. This inference is particularly robust owing to our conservative assumption that the population of whales using Glacier Bay and Icy Strait was closed when, in reality, there is regular within- and among-year mixing of individuals across the larger southeast Alaska region (Hendrix et al., 2012). The regional population of whales would likely serve as buffer for population impacts at the spatial scale of the park, assuming the collision rate in the Glacier Bay region, where whales and ships are highly concentrated (Webb and Gende, 2015), is equal to or higher than other parts of Alaska. Our demographic rates were also relatively conservative, although we assumed that productive oceanographic conditions will persist over the next several decades.

A key (and uncertain) parameter in our analysis was the ship strike rate, which we varied between 2 and 4 collisions per 4622 ship entries, depending upon the historical rate of detection. We highlight that the strike rate parameter was calculated independent of the number of whales available to be struck, which has more than doubled since 1985 (Gabriele et al., 2017). If collisions are a random process, a doubling of the whale population, while holding constant the number of ship transits and the area in which they interact, should, theoretically, result in higher rates of collisions. However, no collisions have been detected in or near the park since 2004 despite ongoing intensive whale monitoring efforts (Gabriele et al., 2017) and the implementation of program in 2006 whereby dedicated observers have boarded over 800 cruise ships (over 40% of all ship entries) with the explicit goal of recording surfacing events of whales around the ships (Gende et al., 2011;

Williams et al., 2016). Thus, detection probability has, if anything, increased coincident with the increases in whales. What's more, our estimated ship strike rate was based on collisions detected through 2009 even though no additional collisions have been detected from 2010 to 2017, a period of time that included > 1600 additional ship visits. The lack of detected ship strikes, despite increased number of whales and increased vigilance suggests that our strike rate may actually overestimate the true collision rate.

Given the low collision rate and multi-decadal growth rates exceeding 4%, it is not surprising then that the number of ship strikes that are likely to occur for MAX ships is 8–56 times below what would need to occur to substantially risk population persistence in the park and surrounding waters. We acknowledge that our model did not account for events or habitat carrying capacity limits (environmental stochasticity; Punt, 2017) which could adversely affect population growth. Nevertheless we hold that the estimated rate of ship strikes should visitation levels be maximized is well below even the most conservative estimates of the number needed to threaten population persistence.

We emphasize that the decision on whether to change the volume of allowable cruise ships into the park is not based solely on the potential impacts to whales. Cruise ships generate pollutants, noise, and create large visual profiles that may disturb marine wildlife or impact other visitors. Thus, in reality, any changes to ship volume will also include consideration of impacts to air quality (Mölders and Gende, 2015), water quality, harbor seals (Young et al., 2014), seabirds (Marcella et al., 2016), and soundscape (Frankel and Gabriele, 2017), among others. While the list of resources and values impacted by ship traffic is diverse, and the relationship between ship volume and impact levels will likely vary among resources, the issue generated using the whale example applies regardless of the resource: at what point does the impact become unacceptable?

4.2. Managing to conserve protected area values

If increases in ship strikes owing to increases in visitation are largely inconsequential to the conservation of whales in Glacier Bay, are managers justified in limiting visitation based on a few additional dead whales over the next three decades to conserve park values? To address this question, it's important to recognize that for the U.S. National Park Service, and many other management agencies, the objectives of the protected areas are to conserve both resources and values. For the NPS, values are defined as encompassing attributes that are both tangible, such as wildlife and pristine water, and intangible, such as "...a park's role in contributions to the national dignity" (NPS, 2006; p 12). Thus the NPS may justifiably limit visitor access if the impact from that level

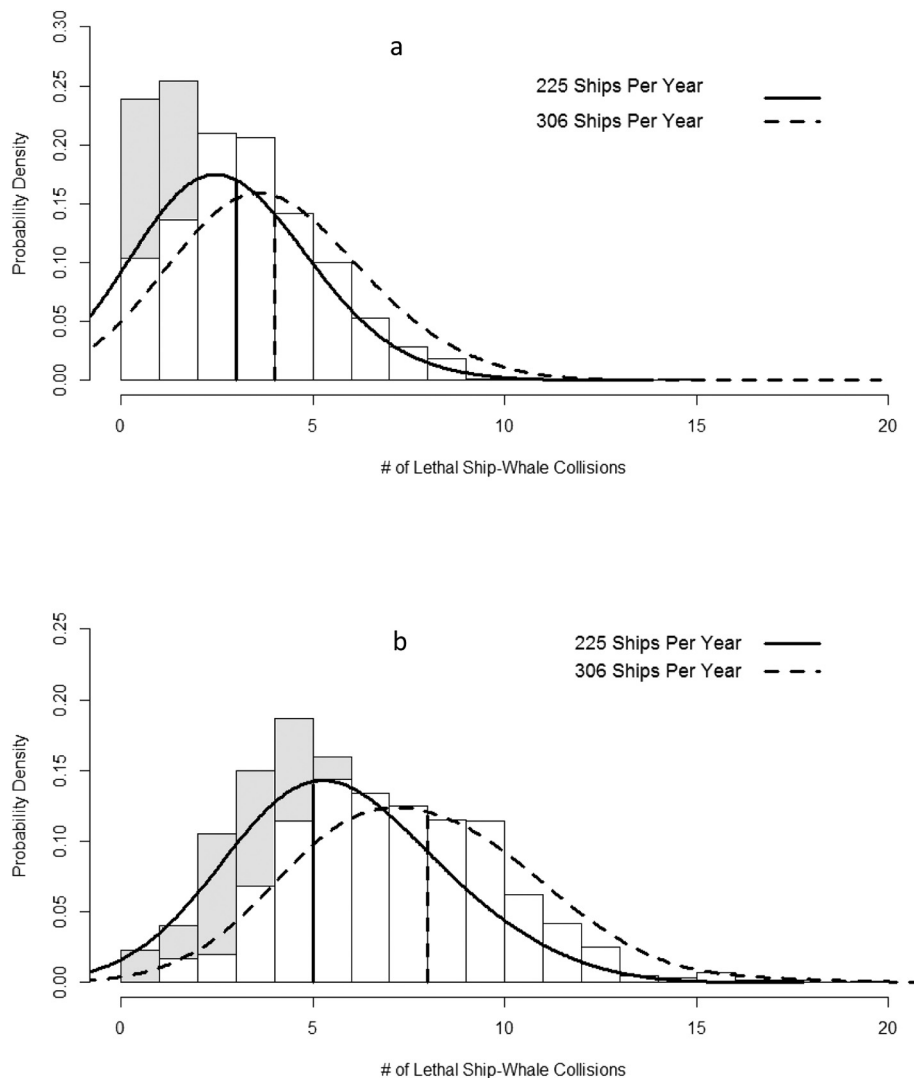


Fig. 3. Probability densities and median estimates (vertical lines) of the number of ship-whale collisions expected to occur in the next 30 years should the annual number of allowable cruise ships into Glacier Bay increase from 225 to 306 assuming historical detection probability was 100% (top) and 50% (bottom).

of visitation is deemed inconsistent with conserving park values, even if that level is far below what is biologically sustainable. Adopting a park value as the conservation target also reconciles the issue that rarely, if ever, is there congruence between the established park boundaries and the biotic boundaries defined by life history, movements, or ecological functions (Newmark, 1985). Indeed, successful conservation of wildlife species should encompass a number of attributes (Redford et al., 2011), which may be impossible to be met by the NPS alone.

Nevertheless, the absence of clearly articulated definitions of park values upon which important management decisions are based, and linkages between those values and metrics that can be measured and monitored, creates nuanced management decisions. In Glacier Bay, providing opportunities for visitors to access and enjoy the park while minimizing the chance of a ship strike is clearly a value, reflected by the suite of programs in place to achieve that goal: dynamic speed restrictions and mid-channel course requirements for vessels and an active outreach effort with marine pilots and bridge personnel emphasizing the importance of whale avoidance, among others. Yet, these efforts reduce, but don't eliminate the chance of a strike. Consequently, should a ship strike (or two) occur, it still places the burden on managers, undoubtedly accompanied by significant public interest, to justify management decisions regarding ship volume. Our study clarifies that a whale death is inconsequential relative to conserving the

resource. The difficulty is identifying the number of whale deaths that threaten park values.

One option is for the NPS and other protected areas to develop a process mirroring that utilized by the U.S. National Marine Fisheries Service (NMFS) to help define acceptable levels of incidental mortality to whales and other marine mammals from fisheries (and ship strikes). The Potential Biological Removal (PBR; Wade, 1998) is calculated by taking the product of the minimum population estimate with half the maximum theoretical net productivity rate (based on maximum rates of increase; Zerbin et al., 2008) and a recovery factor (FR). While the abundance and productivity components make biological sustainability implicit, the recovery factor allows managers to adjust the PBR to meet specific management objectives (Taylor et al., 2003; Cooke et al., 2012). In doing so, PBR allows managers to define acceptable levels of incidental mortality within the range of what's sustainable. In addition to meeting the goals of the U.S. Marine Mammal Protection Act of insuring the population remains (or recovers to) levels between carrying capacity and maximum net productivity level (Cooke et al., 2012), the process also provides a transparent metric that can be monitored and communicated to stakeholders. While PBR is not directly applicable to protected areas owing to population processes, having a PBR-like process to help define, in a consistent and transparent manner, a range of acceptable impact levels within the broader spectrum of what's

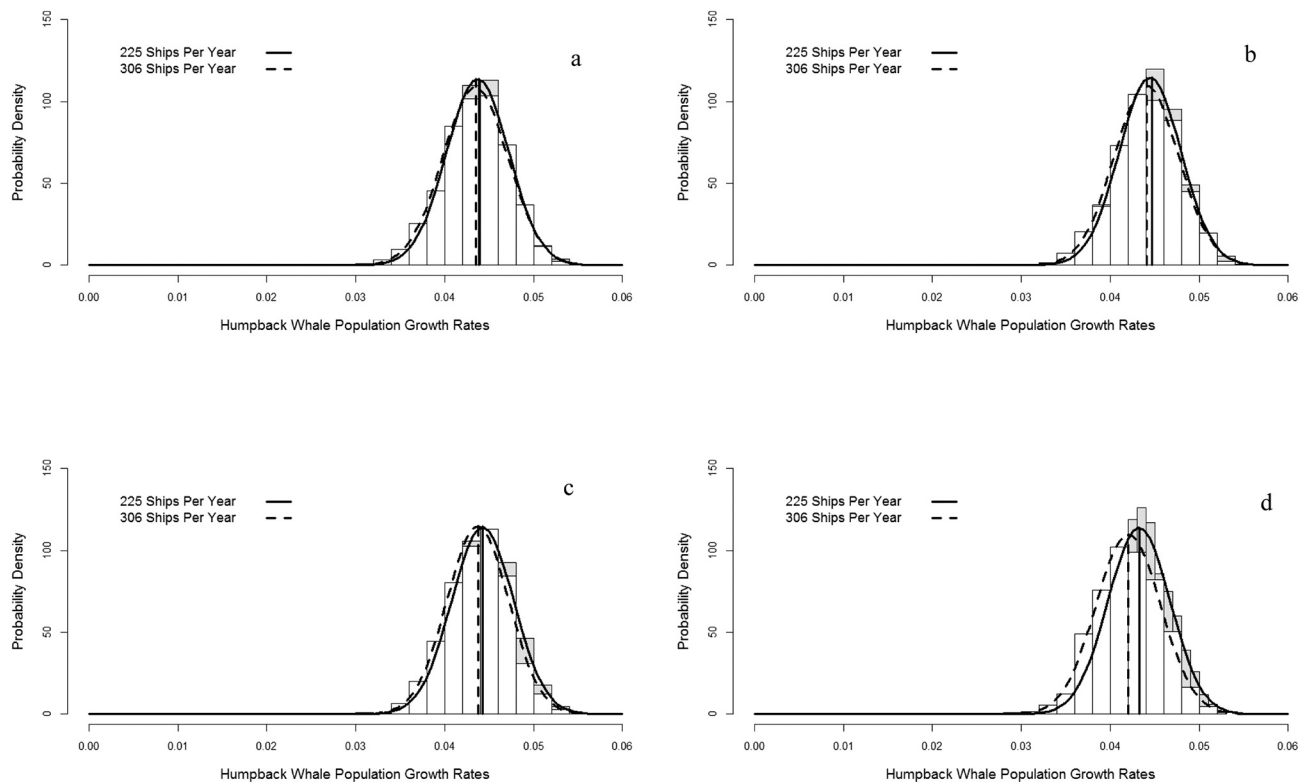


Fig. 4. Probability densities and medians (vertical lines) for parameter estimates of the growth rate of humpback whales using Glacier Bay and surrounding waters for BASE (225) vs. MAX (306) annual cruise ship entry levels assuming historical detection probability of ship strikes was 100% (top 2 panels) or 50% (bottom two panels) and all future collisions will be random with respect to age and sex of whales struck (a, c) or only involve adult females (b, d).

sustainable, may prove beneficial to park managers faced with significant pressure to act following an event (such as a dead whale washing up on the beach), even if that event will have few if any population consequences.

5. Conclusions

Managers of marine and terrestrial protected areas including U.S. national parks and World Heritage Sites are tasked with conserving the site's resources and values. While resources are generally easy to define (e.g., the whale population), defining site values can be more nebulous. Thus site managers are tasked with evaluating the impacts of appropriate uses not just in the context of conservation of the resources but also for conserving site values. We expect that similar intangible values

also apply in Glacier Bay. Even if increased numbers of ship strikes are biologically sustainable, the number of expected dead whales with increased ships may be inconsistent with park values and management goals. The implication is that decisions consistent with ensuring that impacts to a park resource are sustainable biologically may not produce the same decision necessary to conserve site values. Unfortunately values are often unstated or undefined, lending opportunities for confusion and potential conflict from other stakeholders who may view biological sustainability as the only metric for decisions.

There are some established processes that exist that can help protected areas define and link those values to biological metrics, such as defining Limits of Acceptable Change through the development of indicators, standards, and rigorous monitoring (Stankey et al., 1984). A PBR-like metric is another. There are also some instances within the

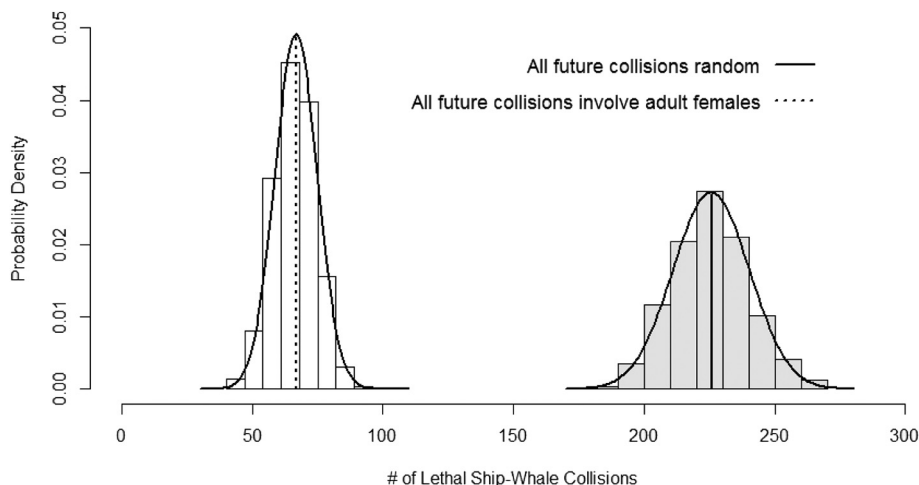


Fig. 5. Median estimates (vertical bars) of the number of ship strikes that would need to occur in Glacier Bay and surrounding waters to force population growth rates from existing levels of 4.4% to 0.2% (no growth). Shaded bars represent a scenario if all future collisions are random with respect to age and sex class; open bars represent scenario where all future collisions involve adult females.

NPS where contentious wildlife decisions were based on conserving park values rather than on biological persistence. For example, in Yukon-Charley Rivers Preserve, continued harvest of a wolf (*Canis lupus*) population measurably impacted by State of Alaska control efforts was determined to be inappropriate, not because the population was at risk of extirpation, but because continued harvest of a depressed population was deemed inconsistent with the core values of the Preserve (see: www.nps.gov/yuch/learn/news/temporary-wolf-closure.htm). This decision was based on a process whereby biological metrics, such as long term wolf survival and the elimination of entire wolf packs from control efforts, were linked to park objectives. Regardless of the process, we encourage the development of clearly stated objectives to support values-based management in all conservation areas where biologically sustainable impacts are not the sole management metric.

Acknowledgements

Reviews of previous drafts by J. Neilson, C. Gabriele, K. Webb, P. Hooe, and L. Welling greatly improved the manuscript. We acknowledge the long-term monitoring data collected by Glacier Bay enabling population estimates and vital rates. We also wish to acknowledge insightful discussion from a session on defining conservation in national parks at the 2015 George Wright Society meeting. Finally, we thank former Glacier Bay National Park superintendent C. Payne, and current Superintendent P. Hooe for their encouragement and institutional support of our results and inferences.

Role of the funding source

This work was financially and institutionally supported by the National Park Service including Glacier Bay National Park and the Central Alaska Network of the NPS Inventory and Monitoring Program. Some of the funding from Glacier Bay was part of the Marine Management Fund, which are funds received from cruise ship passenger fees provided by cruise ship companies as part of their concessions contracts authorized under the Omnibus Parks and Public Lands Management Act of 1996 (Public Law 104-333, Sec. 703). In sum, these funds are allocated solely at the discretion of Glacier Bay management to, in part, "...conduct investigations to quantify any effect of permittees' activity on wildlife and other natural resource values of Glacier Bay National Park. The investigations provided for in this subsection shall be designed to provide information of value to the Secretary, in determining any appropriate limitations on permittees' activity in Glacier Bay". The authors acknowledge no conflict of interest associated with this work including financial, personal or other relationships with other people or organizations within three years of beginning the submitted work that could inappropriately influence, or be perceived to influence, this work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2018.04.038>.

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